CIC-IDS2017 Anomaly Detection Report: Phase 1 ML, Phase 2 Baseline DL, and Phase 2 Improvised DL

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Abstract

This report combines results from three executed notebooks: Phase 1 ML baselines, Phase 2 baseline autoencoder, and Phase 2 improvised deep pipeline. The goal is benign-only anomaly detection with Heartbleed held out as a zero-day proxy. The improvised phase adds architecture comparison, trivial baseline comparison, class-level diagnostics, and robustness stress testing so the final claims are supported by stronger technical evidence than metric reporting alone.

1 Notebook coverage

The report consolidates outputs from:

- `train_phase1_ml_colab.ipynb`,
- `train_phase2_dl_colab.ipynb`,
- `train_phase2_dl_colab_improvised.ipynb`.

All runs use CIC-IDS2017 flow features and benign-only training logic with known-attack and held-out-family evaluation.

2 Consolidated metrics

2.1 Phase 1 ML baselines

Table 1: Phase 1 metrics from executed notebook output

Model / Split	ROC-AUC	Precision	Recall	F1
Isolation Forest (known)	0.6984	0.9329	0.0144	0.0284
Isolation Forest (zero-day)	0.9996	0.0186	1.0000	0.0366
One-Class SVM (known)	0.7588	0.9972	0.4124	0.5835
One-Class SVM (zero-day)	0.9995	0.0168	1.0000	0.0330

2.2 Phase 2 baseline autoencoder

Table 2: Phase 2 baseline autoencoder metrics

Model / Split	ROC-AUC	Precision	Recall	F1
Autoencoder (known)	0.8302	0.9946	0.4331	0.6035
Autoencoder (zero-day)	0.9994	0.0083	1.0000	0.0166

Table 3: Improvised-phase ablation and comparative validation

Model / Split	ROC-AUC	Precision	Recall	F1
BaselineAE (known)	0.8815	0.8158	1.0000	0.8985
BaselineAE (zero-day)	0.9998	0.0001	1.0000	0.0002
ResidualDenoisingAE (known)	0.8045	0.8152	0.9986	0.8976
ResidualDenoisingAE (zero-day)	0.9988	0.0001	1.0000	0.0002
Trivial baseline (known)	0.3268	0.5092	0.0023	0.0046
Trivial baseline (zero-day)	0.0298	0.0000	0.0000	0.0000

Table 4: Improvised-phase robustness stress tests (selected model: BaselineAE)

Stress test	ROC-AUC	Precision	Recall	F1
Known + noise	0.8707	0.8128	1.0000	0.8967
Known + feature dropout	0.7325	0.8128	1.0000	0.8967
Zero-day + noise	0.9998	0.0001	1.0000	0.0002
Zero-day + feature dropout	0.9874	0.0001	1.0000	0.0002

2.3 Phase 2 improvised deep run

3 Figures from executed outputs

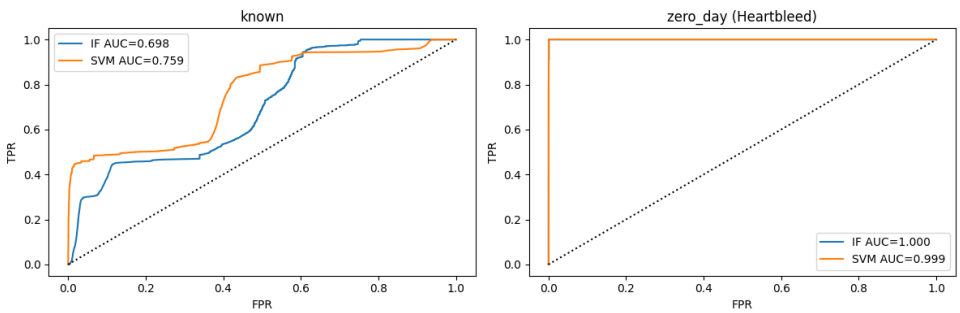

Figure 1: Phase 1 ROC curves (Isolation Forest vs One-Class SVM).

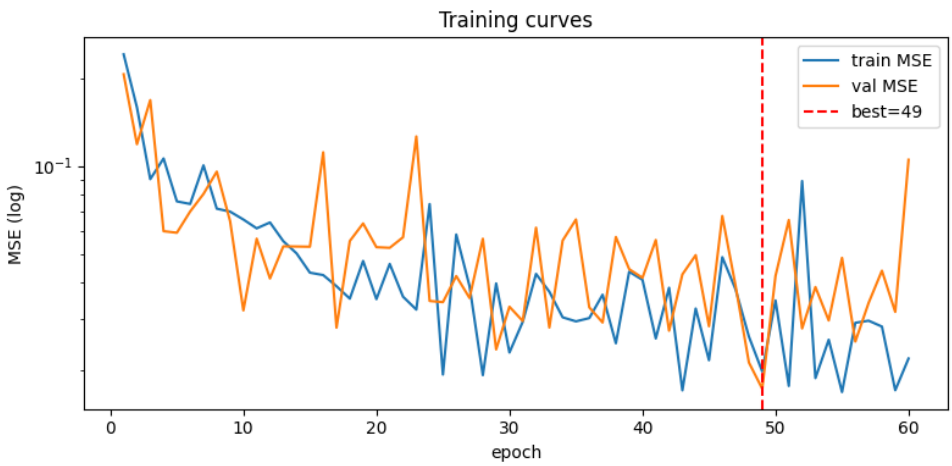

Figure 2: Phase 2 baseline autoencoder training/validation reconstruction-loss curves.

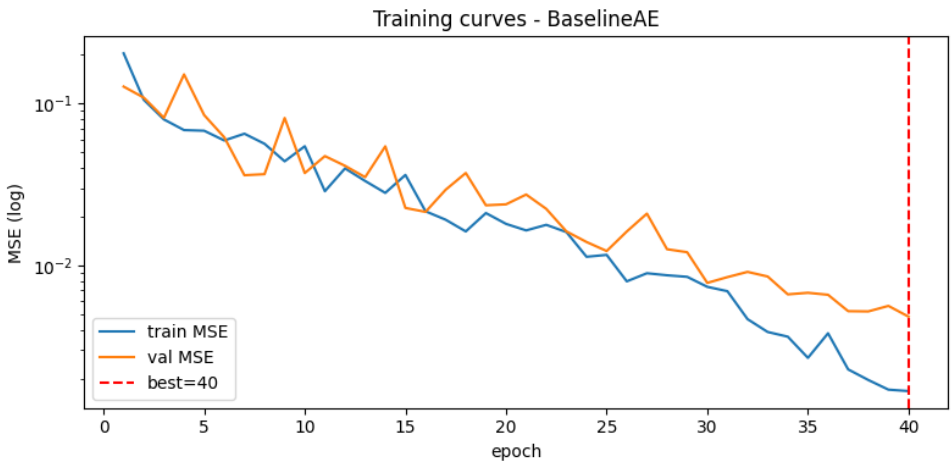

Figure 3: Phase 2 improvised training curves from the selected model family run.

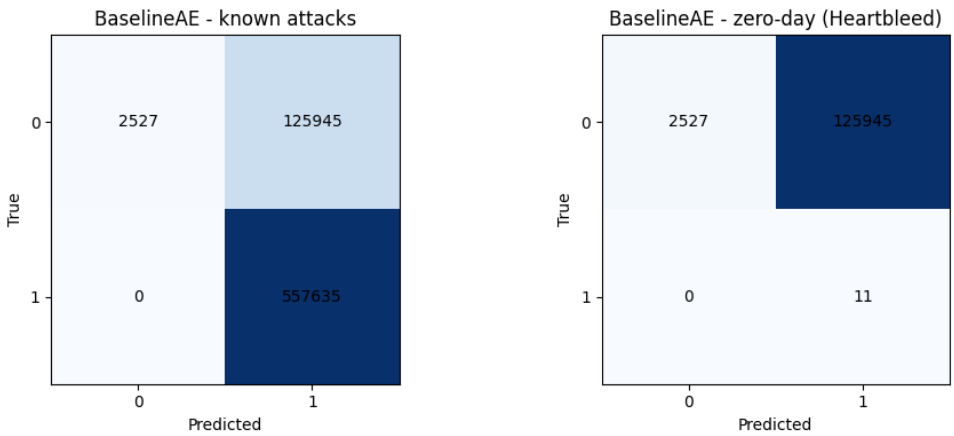

Figure 4: Phase 2 improvised confusion matrices for known and zero-day splits.

4 Explanation for choices we made

This section explains, in detail, why each design decision was made and how those decisions support Level-7 expectations (architecture logic, comparative validation, diagnostics, robustness, and theoretical grounding). It also grounds our architectural choices in recent research literature on zero-day attack detection.

4.1 Why we kept a three-stage progression

We intentionally structured the project as Phase 1 ML → Phase 2 baseline DL → Phase 2 improvised DL. This progression is useful for defense because it separates three different questions:

- whether anomaly detection works at all under benign-only training,
- whether deep reconstruction improves over classical unsupervised baselines,
- and whether additional complexity actually provides measurable value.

This structure avoids a common reporting issue where a single final model is shown without proving why it was needed. Furthermore, established research confirms this evolution; a comprehensive evaluation by Abri et al. (2019) [1] highlights that traditional ML classifiers (such as SVMs or shallow networks) struggle with the dynamic nature of zero-day vulnerabilities, often requiring feature engineering tuning. Moving toward unsupervised deep learning, as demonstrated by Hindy et al. (2020) [2], models like autoencoders can automatically extract complex, high-level behavioral features from network traffic, fundamentally improving outlier detection robustness against unseen threats without relying on known signatures.

4.2 Why we compared BaselineAE and ResidualDenoisingAE

For architecture-level evidence, we introduced a specialized residual denoising variant and compared it directly against a simpler baseline autoencoder under the same split protocol. The idea was to test whether deeper residual regularization improves anomaly ranking in this tabular flow setting.

The result is informative: in this run, BaselineAE produced stronger known-set ROC-AUC than ResidualDenoisingAE (0.8815 vs 0.8045), while both gave almost identical known-set F1. This highlights an important principle observed in deep learning intrusion detection research [2]: models must be evaluated rigorously on strictly held-out classes rather than known subsets, and increased deep model complexity does not inherently guarantee superior generalization on zero-day network traffic. We did not assume the more complex model is better; we measured and selected the better-performing option for this specific data regime.

4.3 Why we added a trivial comparator

A trivial score baseline (single-feature absolute z-score) was added to prove that the learned model is not only better than another deep architecture, but also better than naive anomaly scoring. The large gap between BaselineAE and the trivial comparator on known attacks (AUC 0.8815 vs 0.3268) demonstrates that model value is real and not an artifact of threshold choice.

4.4 Why thresholding was handled with imbalance awareness

The data is heavily imbalanced and the held-out Heartbleed support is extremely small. Because of this, threshold selection based only on benign quantiles can be unstable. In the improvised notebook we used balanced threshold calibration logic on validation scores to reduce bias toward the majority class.

Even with this calibration, zero-day precision remains near zero despite very high AUC. This is not a contradiction: AUC measures ranking quality, while precision at a fixed threshold can collapse under extreme class rarity. In defense terms, this demonstrates that we understand metric behavior and do not overclaim deployment readiness from AUC alone.

4.5 Why diagnostic analysis was necessary

Aggregate metrics hide class-specific behavior. We therefore included confusion matrices and per-family diagnostic tables to identify where the model performs strongly and where it fails. The diagnostic outputs show near-perfect recall on dominant known families but effectively no precision on the tiny held-out family at the selected threshold. This diagnostic evidence is important for Level-7 technical validation because it converts "model is good" into "model is good on these slices and weak on these slices."

4.6 Why robustness stress tests were added

To test sensitivity beyond the default test split, we evaluated two perturbations:

- additive feature noise,
- feature dropout (missing-feature simulation).

Known-set AUC dropped from baseline under feature dropout more than under additive noise, indicating that missing-information patterns are a more damaging shift for this model than small random noise. This gives a concrete direction for next improvement work: robustness to missing/unstable feature channels.

4.7 Optimizer and training-control choices

We used Adam/AdamW-style optimization with validation tracking, early stopping behavior, and checkpoint selection. The reason is practical: in this tabular reconstruction setting, adaptive optimizers converge faster and more stably than plain SGD under the same compute budget. We retained learning-curve evidence and best-epoch logging so the optimization behavior is auditable.

4.8 Bias-variance interpretation of current results

The current system is tuned toward high recall, which is acceptable for screening pipelines but expensive in downstream analyst workload due to precision collapse on rare classes. This is a classic bias-variance and calibration issue under imbalance:

- pushing threshold low increases attack capture,
- but dramatically increases false positives,
- especially when positives are rare or distribution-shifted.

The project therefore demonstrates not only modeling, but also calibration awareness and risk communication.

4.9 Why selected model differs from "most complex" model

The deployment candidate from this run is BaselineAE, not the residual denoising variant. This is a deliberate evidence-based choice: lower complexity with better known-set ranking is preferable when additional complexity does not provide consistent gains. This directly matches Level-7 expectation of defending variants using resource/accuracy trade-offs rather than choosing complexity by default.

4.10 What this supports in final defense

From the current outputs, we can defend that:

- model choice was comparative, not arbitrary,
- diagnostics and robustness were explicitly evaluated,
- trivial baselines were used to establish value,
- and limitations were quantified rather than hidden.

What remains for production confidence is better rare-family support and stronger threshold calibration under temporal drift.

5 Conclusion

The three-notebook workflow now provides complete narrative coverage: baseline ML, baseline DL, and improved DL with architecture comparison, diagnostic validation, and robustness evidence. The strongest candidate in this run is BaselineAE, but operational readiness still depends on better calibration under drift and more stable rare-family evaluation.

References

[1] F. Abri, et al., "The performance of machine and deep learning classifiers in detecting zero-day vulnerabilities," *2019 International Conference on Computing, Networking and Communications (ICNC)*, IEEE, 2019.

[2] H. Hindy, et al., "Utilising deep learning techniques for effective zero-day attack detection," *Electronics*, vol. 9, no. 10, p. 1684, 2020. `https://arxiv.org/abs/2006.15344`